Reproduction

20 questions: 7 multiple choice, 13 structured.

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1 · Part 1 Paper 1 & 2 Reproduction [1]

Which of these is an example of asexual reproduction?

A a sperm cell fusing with an egg cell

B a bacterial cell dividing to form two genetically identical cells

C a bird laying a fertilised egg

D a male fish and a female fish releasing their gametes into the water

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2 · Part 1 Paper 1 & 2 Reproduction [1]

What is formed when a male gamete fuses with a female gamete during fertilisation?

A an embryo

B a fetus

C an ovum

D a zygote

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3 · Part 1 Paper 1 & 2 Reproduction [1]

Which part of a flower produces pollen grains?

A anther

B ovary

C stigma

D style

4 · Part 1 Paper 1 & 2 Reproduction [1]

After fertilisation in a flowering plant, which part of the flower develops into a seed?

A ovary

B stigma

C ovule

D anther

5 · Part 1 Paper 1 & 2 Reproduction [1]

In the human female reproductive system, where does fertilisation normally take place?

A ovary

B uterus

C cervix

D oviduct

6 · Part 1 Paper 1 & 2 Reproduction [1]

Blood tests show that a woman has had a high concentration of progesterone in her blood for the last eight weeks. She has not menstruated during this time.

What is the most likely explanation?

A She is pregnant.

B She is about to ovulate.

C Her uterus lining is breaking down.

D Her ovaries have stopped producing hormones.

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7 · Part 1

Paper 2 only

Reproduction

[1]

Which hormone stimulates the ovaries to produce oestrogen during the menstrual cycle?

A LH

B FSH

C progesterone

D testosterone

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8 · Part 1

Paper 1 & 2 Reproduction [6]

Strawberry plants grow long side shoots called runners along the surface of the soil. New plants grow where the runners touch the soil. Gardeners also produce new geranium plants by taking cuttings.

(a) Producing new plants from runners is a natural method of asexual reproduction. Give two differences between asexual reproduction and sexual reproduction. [2]

(b) Describe how a gardener could produce new geranium plants by taking cuttings. [2]

(c) A farmer has a strawberry plant that produces very sweet fruit. She uses runners from this plant, rather than its seeds, to produce new plants.

Explain one advantage and one disadvantage to the farmer of producing new plants from runners rather than from seeds. [2]

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9 · Part 1

Paper 1 & 2 Reproduction [6]

The table describes the flowers of two plant species, P and Q.

Feature	Flower P	Flower Q
Petals	small and green	large and brightly coloured
Scent and nectar	no scent and no nectar	strong scent and produces nectar
Anthers	hang outside the flower on long filaments	inside the flower, firmly attached
Stigma	large and feathery, hangs outside the flower	small and sticky, inside the flower
Pollen grains	small, light and smooth; produced in very large numbers	larger and spiky; produced in smaller numbers

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(a) Deduce which flower, P or Q, is wind-pollinated. [1]

(b) Explain how the anthers and the stigma of flower P adapt it for its method of pollination. [4]

(c) Explain why flower P produces very large numbers of pollen grains. [1]

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10 · Part 1

Paper 1 & 2 | Reproduction | [7]

When a bee visits a flower to feed on nectar, pollen grains stick to its body. The bee then visits another flower of the same species, and some of the pollen grains are transferred to the stigma of this second flower.

(a) The pollen grains of many insect-pollinated flowers have a sticky or spiky surface. Explain how this helps pollination. [2]

(b) Describe how fertilisation takes place after a pollen grain lands on the stigma. [3]

(c) State what the ovary develops into after fertilisation. [1]

(d) The seeds produced by this plant are genetically different from each other. Explain why. [1]

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11 · Part 1

Paper 1 & 2 Reproduction [7]

Kofi investigated the conditions needed for the germination of cress seeds. He set up four test tubes. Each tube contained 20 cress seeds on cotton wool. All four tubes were kept in the dark. After 5 days he counted the number of seeds that had germinated in each tube.

Tube	Conditions	Number of seeds germinated after 5 days
1	dry cotton wool, 20 °C	0
2	moist cotton wool, 20 °C	19
3	moist cotton wool, in a refrigerator at 4 °C	2
4	cotton wool soaked in water that had been boiled and cooled, covered with a layer of oil, 20 °C	0

(a) Calculate the percentage of seeds that germinated in tube 2.

Use the formula: percentage germinated = (number of seeds germinated ÷ total number of seeds) × 100 [2]

(b) Identify the two tubes that Kofi should compare to show that seeds need water to germinate. [1]

(c) Explain why no seeds germinated in tube 4. [3]

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(d) Give one way that Kofi could make his results more reliable. [1]

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12 · Part 1 Paper 1 & 2 Reproduction [7]

Lena wants to investigate how temperature affects the germination of mung bean seeds. She has a large supply of mung bean seeds, Petri dishes, cotton wool, a measuring cylinder, water and several incubators that can each be set to a different temperature.

(a) State a suitable dependent variable for Lena to measure. [1]

(b) Design an investigation that Lena could carry out to find the effect of temperature on the germination of mung bean seeds. [6]

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13 · Part 1

Paper 1 & 2 Reproduction [9]

Mei planted 50 bean seeds in moist soil in a warm, sunny place. Every two days, starting on day 0, she removed five of the seeds or seedlings, dried them in an oven and found their mean dry mass. The first green leaves appeared above the soil on day 6.

Day	Mean dry mass of one seedling / g
0	0.80
2	0.76
4	0.70
6	0.64
8	0.66
10	0.74
12	0.86

(a) Describe the changes in the mean dry mass of one seedling between day 0 and day 12. [2]

(b) Calculate the percentage decrease in the mean dry mass of one seedling between day 0 and day 6.

Use the formula: percentage change = (change in mass ÷ starting mass) × 100 [2]

(c) Explain why the dry mass of the seedlings decreased between day 0 and day 6. [3]

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(d) Explain why the dry mass of the seedlings increased after day 6. [2]

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14 · Part 1 Paper 1 & 2 Reproduction [6]

The table shows some structures in the human reproductive systems and their functions.

(a) Complete the table by giving the missing structures and functions.

Structure	Function
testis	(i) .....
(ii)	carries sperm from the testis to the urethra
prostate gland	(iii) .....
(iv)	where sperm are deposited during sexual intercourse

[4]

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(b) The wall of the uterus contains a thick layer of muscle. Suggest why this is important. [1]

(c) The testes are held in the scrotum, outside the main body cavity. Explain why. [1]

15 · Part 1

Paper 1 & 2 Reproduction [7]

The table shows the concentrations of oestrogen and progesterone in the blood of a woman during one 28-day menstrual cycle. Menstruation started on day 1.

Day	Oestrogen concentration / arbitrary units	Progesterone concentration / arbitrary units
1	10	2
4	12	2
8	20	2
12	45	3
14	30	4
18	22	12
21	25	18
25	15	10
28	10	3

(a) Identify the day on which the oestrogen concentration was highest. [1]

(b) Calculate the percentage increase in oestrogen concentration between day 4 and day 12.

Use the formula: percentage change = (change in concentration ÷ starting concentration) × 100 [2]

(c) Describe the role of oestrogen in the menstrual cycle. [2]

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(d) Explain why menstruation starts again at about day 28. Use information from the table in your answer. [2]

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16 · Part 1

Paper 2 only

Reproduction

[7]

Amara has been unable to become pregnant. Her body releases very little FSH, and she does not ovulate. A doctor treats her with daily injections of FSH.

Each egg develops inside a fluid-filled follicle in the ovary. The doctor uses scans to measure the diameter of the largest follicle in Amara's ovaries. When this follicle reaches a diameter of 18 mm, the doctor gives Amara a single injection of LH.

Day of treatment	Diameter of largest follicle / mm
1	4
3	7
5	11
7	15
9	18

(a) Explain why the injections of FSH can help Amara to become pregnant. [2]

(b) Calculate the mean rate of increase in the diameter of the largest follicle between day 1 and day 9.

Use the formula: rate = change in diameter ÷ time taken [2]

(c) Explain why the doctor gives the injection of LH when the follicle reaches a diameter of 18 mm. [2]

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(d) If the dose of FSH is too high, several eggs may mature at the same time. Suggest one risk of this for Amara. [1]

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17 · Part 1

Paper 2 only

Reproduction

[8]

The table shows the concentrations of four hormones in the blood of a woman during one 28-day menstrual cycle. All concentrations are in arbitrary units.

Day	FSH	LH	Oestrogen	Progesterone
1	8	4	9	1
5	10	5	15	1
9	7	6	30	1
12	9	20	52	2
14	14	60	35	3
16	6	12	22	10
21	3	4	26	20
26	5	4	11	6

(a) Identify the day on which ovulation most likely took place. Give a reason for your answer. [2]

(b) Explain how the changes in the concentrations of FSH, LH and oestrogen between day 1 and day 14 lead to ovulation. [4]

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(c) Calculate how many times greater the LH concentration was on day 14 than on day 9.
Show your working. [2]

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18 · Part 1

Paper 2 only

Reproduction

[6]

The combined contraceptive pill contains oestrogen and progesterone. A woman takes one pill each day for 21 days and then takes no pills for 7 days. While she is taking the pills, the concentrations of oestrogen and progesterone in her blood stay high.

(a) Explain how taking the pill prevents pregnancy. [3]

(b) Suggest why it is important that the woman takes her pill at about the same time every day. [2]

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(c) The table shows the number of pregnancies in one year for every 100 women using the pill.

Way the pill is used	Number of pregnancies per 100 women in one year
perfect use (every pill taken exactly as instructed)	0.3
typical use (how most women actually use it)	9

Suggest why the number of pregnancies is higher with typical use. [1]

19 · Part 1

Paper 1 & 2 Reproduction [7]

After fertilisation, the fertilised egg cell travels to the uterus and becomes attached to the uterus lining. It develops into an embryo. The embryo is surrounded by amniotic fluid and is connected to a placenta.

(a) Describe how the fertilised egg cell develops into an embryo. [2]

(b) Describe the role of the placenta in the nutrition of the developing embryo. [3]

(c) Explain how the amniotic fluid protects the developing embryo. [2]

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20 · Part 1 Paper 1 & 2 Reproduction [7]

During puberty, hormones cause secondary sexual characteristics to develop in boys and girls.

(a) State the name of the hormone that causes secondary sexual characteristics to develop in boys. [1]

(b) Describe two secondary sexual characteristics that develop in boys during puberty. [2]

(c) Describe two secondary sexual characteristics that oestrogen causes to develop in girls during puberty. [2]

(d) Some female athletes have illegally used drugs that have the same effects as the hormone named in part (a).

Suggest two effects these drugs could have on the body of a female athlete. [2]

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